2025 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW) | 979-8-3315-1931-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICASSPW65056.2025.11011079

Hybrid Encoding-based Quantum Self Attention for Financial Time Series Analysis

Lokes S
Wells Fargo
Bengaluru, India
Lokes.S@wellsfargo.com

Suresh Reddy
Wells Fargo
Hyderabad, India
suresh.reddy@wellsfargo.com

Samuel Yen Chi Chen
Wells Fargo
New York, NY, USA
yen-chi.chen@wellsfargo.com

***Abstract*—Self-attention mechanisms, central to Transformer architectures, have become foundational in the development of large language models (LLMs) and various advanced machine learning systems. These mechanisms are pivotal for effectively capturing complex data dependencies and relationships, which is crucial for tasks such as natural language understanding and generation. However, as dimensionality increases, self-attention mechanisms face significant challenges related to computational efficiency and scalability. This paper introduces an encoding-based quantum self-attention mechanism designed to address these issues by embedding classical data into quantum states. This approach facilitates more efficient computation and reduces model complexity. To further enhance performance for real-world applications, particularly in financial time series forecasting, we propose a hybrid model that integrates Long Short-Term Memory (LSTM) networks with quantum self-attention mechanism. Given the constraints of the Noisy Intermediate-Scale Quantum (NISQ) era, this hybrid approach represents a plausible solution, leveraging the strengths of both classical and quantum methods and focusing on achieving faster convergence. Experimental results demonstrate that our approach converges significantly quicker than traditional models, showcasing the advantages of integrating quantum techniques with classical machine learning for practical applications.**

***Index Terms*—Self-Attention, Transformers, Quantum Self-Attention, LSTM, Quantum Neural Networks**

I. Introduction

Quantum computing is revolutionizing the approach to complex computational tasks, offering capabilities that surpass classical computing through the quantum principles [1], [2]. This emerging technology holds significant potential to transform various fields, particularly machine learning, where quantum algorithms could dramatically improve both efficiency and accuracy [3]–[6].

Despite these advancements, traditional machine learning models, especially self-attention mechanisms, encounter substantial challenges in scaling and processing high-dimensional data. These issues are particularly evident in tasks that require detecting intricate dependencies within data, such as in time series analysis [7]. To address these limitations, this paper introduces an encoding-based architecture for quantum self-attention mechanisms. By embedding classical data into quantum states and utilizing quantum circuits, the proposed approach facilitates efficient self-attention computation, significantly reducing complexity while enhancing the model's ability to capture intricate patterns within the data [8].

Given the current constraints of quantum computing, particularly in the Noisy Intermediate-Scale Quantum (NISQ) era, this paper supports a hybrid approach. The proposed model integrates a Long Short-Term Memory (LSTM) network with a quantum self-attention mechanism to leverage the strengths of both classical and quantum computing. This hybrid architecture is specifically designed to detect precise volatility in time series data, which is crucial for financial forecasting and other practical applications [9]. The results reveal that this approach converges significantly faster than traditional models and thereby enhances the performance compared to traditional methods, highlighting the potential of quantum-enhanced machine learning.

II. Related Works

Quantum Long Short-Term Memory (QLSTM) and Quantum Recurrent Neural Networks (QRNNs) have made strides in quantum machine learning for sequential data. [10] introduced QLSTM, which adapts the classical LSTM architecture to quantum circuits, showing its ability to capture long-term dependencies more efficiently with fewer parameter. [11] explored QRNNs, which encode temporal dependencies using quantum states. Hybrid Quantum-Classical Recurrent Networks (QCRNs) [12] combine quantum and classical layers to perform well under NISQ constraints.

Despite these advances, these models struggle with complex dependencies such as natural language processing, leading to the development of Quantum Self-Attention Mechanisms, which excel at capturing intricate data relationships. Quantum Self-Attention Neural Networks (QSANN) use Gaussian Projection Quantum Self-Attention (GPQSA) to efficiently map high-dimensional quantum states into classical space, enhancing NLP tasks [13]. QSAM, optimized for NISQ devices, outperforms classical self-attention models in text categorization [8].

Integrating Quantum Logical Similarity (QLS) and the Quantum Bit Self-Attention Score Matrix (QBSASM) im-

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Code available at: https://github.com/lokesMCI/QSAN_with_LSTM

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proves quantum data representation and learning speed [14], while Quantum Kernel Self-Attention Networks (QKSAN) leverage quantum kernels for handling high-dimensional data [15]. These advancements demonstrate quantum self-attention's potential to enhance performance across various machine learning applications.

III. Variational Quantum Circuits

Variational Quantum Circuits (VQCs) play a crucial role in the development of hybrid quantum-classical algorithms, particularly in the Noisy Intermediate-Scale Quantum (NISQ) era, where quantum hardware is limited by noise and the number of qubits. VQCs combine quantum circuits with classical optimization techniques to minimize a cost function iteratively, allowing quantum systems to solve complex problems even with current hardware limitations [16], [17].

In a VQC, quantum gates are parameterized, and these parameters are adjusted or trained through a classical optimizer, such as gradient descent. The quantum circuit prepares a quantum state, which is measured, and the outcomes are used to calculate the cost function. This hybrid loop enables VQCs to solve various tasks quantum machine learning [18] such as classification [19]–[21], sequence learning [10], natural language processing [22]–[25] and reinforcement learning [26]–[30]. VQCs are also good at tasks like optimization and chemistry simulations. One of the key advantages of VQCs is their flexibility—they can be tailored to specific problem types, making them applicable in various fields, such as time series analysis and quantum-enhanced machine learning [31], [32].

IV. Self Attention Mechanism

Self-attention is a key component of transformer models, enabling them to capture complex dependencies in data, essential for tasks like sequential learning and time series analysis [7], [33]. Unlike RNNs like LSTM, self-attention allows for direct interaction between all elements in a sequence, enhancing the ability to model long-range relationships and improving efficiency through parallel processing [34], [35].

Self-attention works by transforming input data into Query (Q), Key (K), and Value (V) vectors and calculating attention scores using a scaled dot product:

$$\text{Attention Score} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)$$

and d_k is a scaling factor used to reduce the magnitude of dot products. This approach excels at handling complex patterns over time, making it superior to traditional RNNs for time series tasks. However, its computational cost scales quadratically with sequence length, leading to research into more efficient variants like sparse attention [36], [37]. Despite these challenges, self-attention remains a crucial tool for advanced data modeling [7], [35].

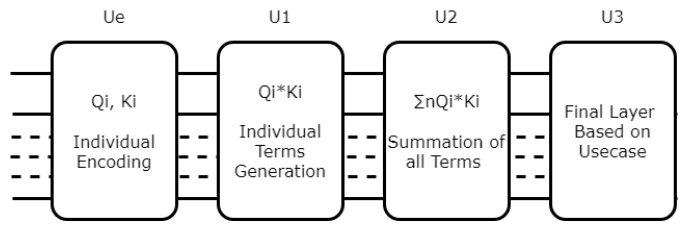

Fig. 1. **Encoding Based Quantum Self Attention**

V. Encoding Based Quantum Self Attention Mechanism

There are various approaches to implementing quantum self-attention mechanisms, and among them, the quantum encoding-based self-attention mechanism is particularly promising. This method leverages quantum computing to enhance self-attention models by embedding classical data into quantum states, allowing for the efficient capture of complex relationships [8].

QSAM employs an angle encoding strategy to transform classical data into quantum states. Specifically, the encoding unit U_e utilizes angle encoding [8], where classical inputs are mapped to rotational angles of qubits. This encoding generates the query Q_i, key K_i, and value V_i vectors for quantum processing [8].

Once the Q_i, K_i, and V_i vectors are encoded, their interactions are computed using controlled rotational gates, optimizing the attention calculations.

1. Calculating $Q_i \cdot K_i \cdot V_i$ with U_1 Circuits: The interaction term $Q_i \cdot K_i \cdot V_i$ is calculated using the first set of quantum circuits, denoted as U_1, which is constructed using controlled rotational gates (e.g., controlled RX gates). These controlled operations efficiently perform the interaction between the quantum states [8]:

$$U_1(Q_i, K_i, V_i) = Q_i \cdot K_i \cdot V_i$$

2. Summing the Weighted Vectors with U_2 Circuits: The second layer of circuits, denoted as U_2, sums the weighted vectors. Here, the summation is performed by applying controlled parameterized gates in a circular fashion on the above generated terms:

$$U_2 \sum_{j=1}^{N} \sum_{i=1}^{N} Q_j \cdot K_i \cdot V_i = \sum_{j=1}^{N} \sum_{i=1}^{N} U_1(Q_j, K_i, V_i)$$

This process ensures that the interactions between different Q_i, K_i, and V_i combinations are efficiently aggregated, enhancing the model's attention mechanism. [8]

3. Final Transformation with U_3 Circuits: The final stage involves processing the aggregated attention scores through the U_3 circuits. These circuits, designed as Variational Quantum Circuits (VQCs), perform the final task-specific transformations and classifications. The VQCs allow for flexible optimization, making it possible to adapt the circuit parameters to the specific requirements of the attention mechanism [8] .

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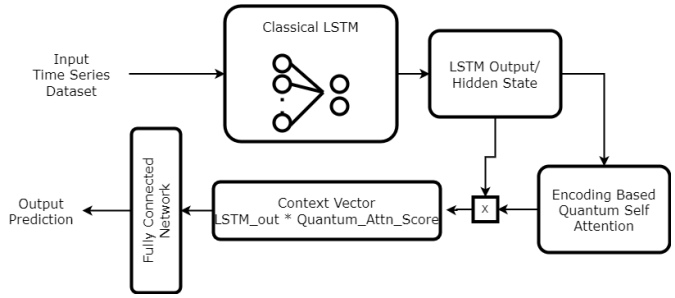

Fig. 2. **Hybrid LSTM Quantum Self Attention Architecture**

VI. Hybrid Quantum Self Attention for time series analysis

Quantum computing has shown remarkable potential, but practical implementations face constraints due to the limitations of near-term intermediate-scale quantum (NISQ) devices. These devices are characterized by limited qubit counts and noisy operations, which pose challenges for scaling up quantum circuits to handle complex tasks efficiently [2], [3]. To overcome these constraints and harness the benefits of quantum computing, hybrid approaches that integrate classical and quantum methods are gaining traction.

One promising hybrid approach involves combining Long Short-Term Memory (LSTM) networks with quantum self-attention mechanisms. LSTM networks are well-suited for time series analysis due to their ability to capture temporal dependencies and patterns within sequential data. They excel at modeling long-term dependencies and handling variable-length time series inputs, making them a staple in time series forecasting and analysis [38]. However, LSTMs may struggle with fine-grained analysis of subtle volatility and short-term fluctuations within the data.

To address this limitation, we integrate the proposed encoding based quantum self-attention mechanism, into the LSTM framework. After the LSTM network processes the data, the quantum self-attention mechanism is applied to refine the analysis further. The quantum self-attention mechanism enhances the model's ability to capture intricate relationships and subtle volatility in the time series data by leveraging quantum circuits to perform attention operations with high precision [8].

In this hybrid framework, time series data is first passed through an LSTM network, which processes the input data and produces a hidden state, representing the LSTM's output. This assists in capturing long-term dependencies and trends [38]. This LSTM output is crucial in computing the quantum self-attention score. Specifically, the hidden state from the LSTM is used as the basis to calculate the quantum self-attention score. This step enables fine-grained attention and analysis of subtle volatility within the time series [8]. This score is then multiplied by the original LSTM output, forming a context vector. The context vector captures both long-term dependencies and fine-grained temporal features within the data. In the final step, the context vector is passed through a fully connected layer, which uses this information to predict the output.

VII. Experiments

In this study, we utilize open-source tools for simulation, specifically PennyLane [39] for designing quantum circuits and PyTorch for constructing the hybrid quantum-classical model.

A. Classification problem

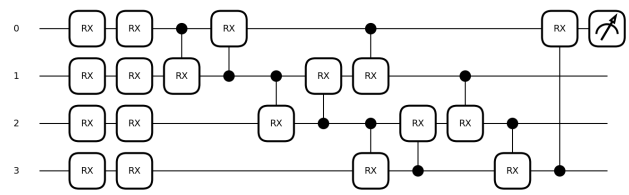

Fig. 3. **Encoding Based Quantum Self Attention circuit diagram**

For our experiments, we utilized the Iris dataset, a widely recognized benchmark in classification tasks. The Iris dataset includes 150 samples of iris flowers, each described by four features: sepal length, sepal width, petal length, and petal width. This dataset's dimensionality is four, corresponding to the size of each feature vector. The manageable size of the Iris dataset allows us to focus on assessing the fundamental performance of our circuit without the additional complexity of larger datasets and facilitates a straightforward comparison with Variational Quantum Circuits (VQC) and Encoding-based Quantum Self Attention mechanism architectures to determine the effectiveness and potential advantages of our proposed approach.

In the context of the Iris dataset, the process begins with encoding the four-dimensional feature vectors into quantum states. This encoding is followed by the application of U_1, U_2, and U_3 circuits, as detailed previously. The U_1 circuits handle the computation of weighted value vectors, U_2 circuits aggregate these outputs to produce attention scores, The detailed implementation of these circuits is illustrated in the circuit diagrams provided. Finally U_3 circuits, designed as VQCs, apply final transformations to produce classification results.

B. Financial Time series data

In our experiments with real-world time series data, we utilized stock price data sourced from Yahoo Finance, focusing on predicting the closing prices of a specific stock. This dataset provides a rich context for evaluating the performance of our hybrid LSTM with Encoding-based Quantum Self-Attention Network. The LSTM network, optimized using Adam with a learning rate of 0.01 and a hidden state size of 4, with a layer depth of 1, processes the sequential stock price data. The Quantum Self-Attention Mechanism, which employs a Nestero Momentum Optimizer from Pennylane [39], refines the LSTM output. Finally, the context vector obtained from attention mechanism is passed through a fully connected network for prediction. This approach aims to address the challenges of forecasting in dynamic financial markets by combining sequential learning with quantum-enhanced attention mechanisms.

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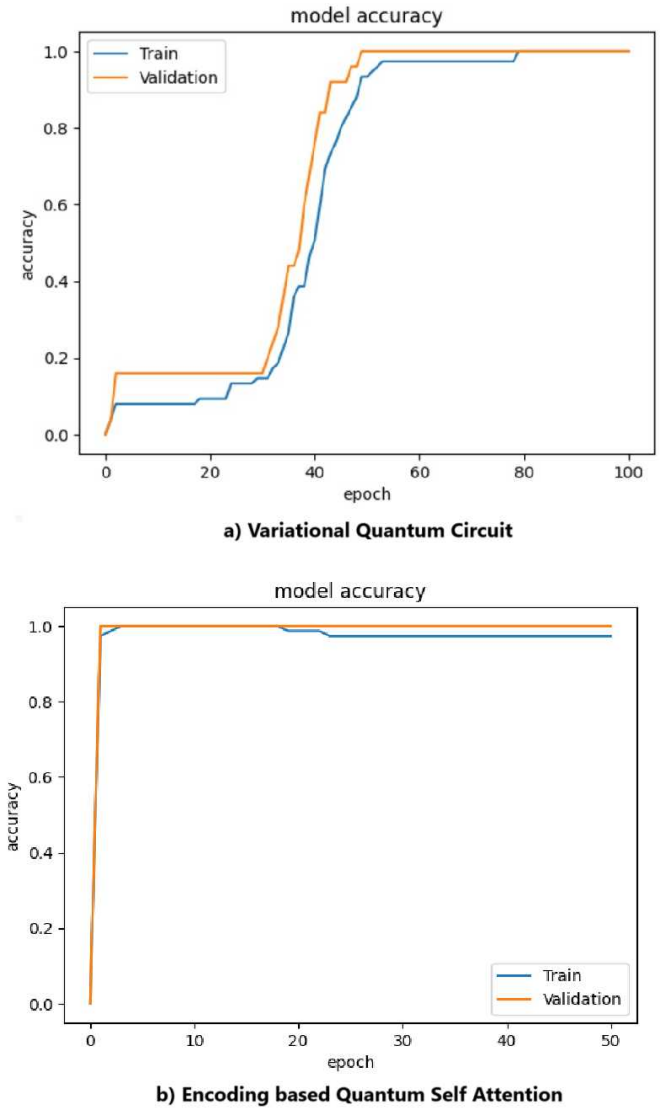

Fig. 4. **Results on IRIS Dataset**

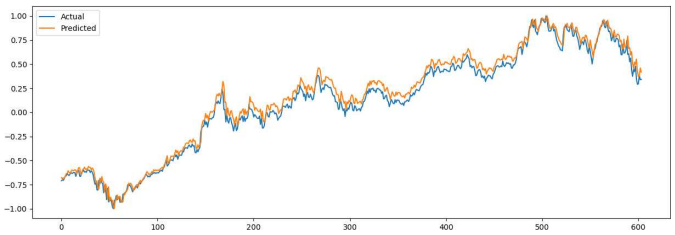

Fig. 5. **Training predictions - Actual vs predicted time series with Hybrid LSTM and Encoding based Quantum Self Attention architecture**

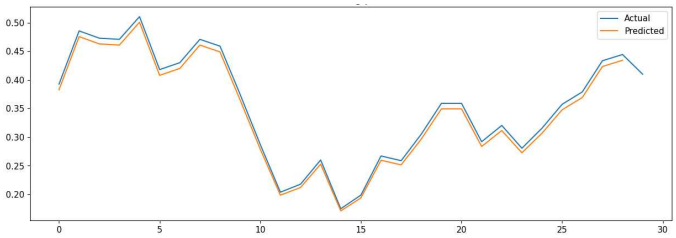

Fig. 6. **Testing predictions - Actual vs predicted time series with Hybrid LSTM and Encoding based Quantum Self Attention architecture**

VIII. Results and Discussion

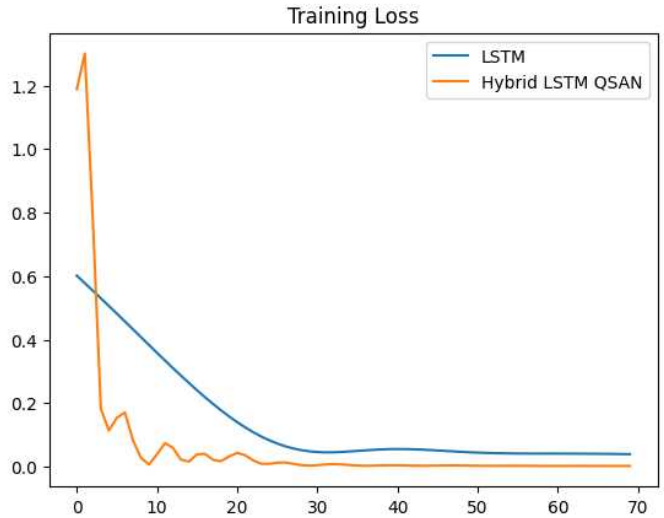

Fig. 7. **Convergence comparison of LSTM vs Hybrid architecture**

To test the base architecture of our Encoding-based Quantum Self-Attention, we first applied it to the Iris classification task. We compared its performance against traditional Variational Quantum Circuits (VQC) and classical neural network architectures. The results demonstrated that our encoding-based quantum self-attention architecture achieved accuracy comparable to classical neural networks while offering the advantage of fewer parameters. Additionally, our Encoding-based self attention architecture exhibited faster convergence compared to the VQC approach, as illustrated in the accompanying figure.

In the context of time series analysis using stock price data, we evaluated our hybrid LSTM and encoding-based quantum self-attention architecture. This hybrid model was compared with classical LSTM networks and traditional self-attention mechanisms. Our hybrid approach achieved convergence significantly earlier compared to classical approach. This improvement in convergence is depicted in the provided figure, highlighting the effectiveness of integrating LSTM with quantum self-attention for handling real-world time series data.

IX. Conclusion

Our study highlights the significant advantages of the hybrid approach combining Long Short-Term Memory (LSTM) networks with the Encoding-based Quantum Self-Attention Mechanism. This hybrid model effectively enhances time series forecasting by integrating the sequential learning strengths of LSTM with the refined attention capabilities of Encoding-based Quantum Self attention mechanism. The results demonstrate that this approach not only improves the model's ability to capture complex temporal dependencies but also achieves faster convergence compared to traditional methods. These findings underscore that the hybridization of quantum self-attention with classical techniques represents a compelling advantage, showcasing the potential of this innovative approach to revolutionize machine learning and forecasting tasks.

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